

The Unreasonable Effectiveness of Quarto

John Paul Helveston¹

¹ George Washington University, Washington D.C., USA

Categories

Best practices or advice to practitioners

Presentation type

Presentation - 30 minutes (including 5-10 minutes for audience questions)

Summary

Quarto markets itself as a publishing framework. But at its core, Quarto is an abstraction layer separating *content* from *format* — a distinction that turns out to be remarkably powerful in the era of AI agents. When an LLM generates a document, content and structure emerge together in a single probabilistic pass, but nothing in that output is guaranteed. Quarto's code-at-render-time model solves this by letting the LLM build the structure while allowing the content that must be exact to be rendered deterministically from data files. The result is reproducible and auditable output that stays correct as your content changes, without needing to re-prompt models. This talk brings the agent + Quarto workflow to life with concrete examples, from a simple one-page PDF to a fully interactive multi-page dashboard integrating JavaScript libraries, Shinylive, and React, showing how asking an LLM for a qmd file instead of a finished document dramatically reduces token overhead while producing more dynamic, trustworthy, and maintainable output.

Target audience for this presentation

Anyone working with agentic workflows to analyze data, write reports, and most other types of work where outputs from LLMs need to be reproducible.

Key Goals or Takeaways

For people to learn about Quarto in general and how to leverage it as an intermediate layer between agents and users to develop reproducible outputs from agentic systems.

How technical is your talk?

Minimally technical, approachable for all researchers

Usage of Large Language Model (LLM) AI systems

Yes, I used LLMs in STUDY PLANNING (or plan to), Yes, I used LLMs when writing CODE (or plan to), Yes, I used LLMs when WRITING or EDITING the presentation (or plan to)